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William Maddox

Senior .NET & React Full Stack Developer

Cloud-native Full-Stack Engineer with 10+ years building distributed, high-availability platforms using ASP.NET Core, C#, React/Next.js, Angular, TypeScript, and event-driven architectures. Designed microservices with Kafka, Cassandra, Redis, SignalR, and AWS/Azure infrastructure, delivering real-time systems at enterprise scale. Proven track record of optimizing performance (40%+ latency reduction), scaling traffic 50x, and leading complex migrations and greenfield builds from concept to production. Combines strong product thinking with deep backend and DevOps expertise.

WORK EXPERIENCE

Senior Full Stack Software Engineer (.NET/React)

May 2022 - Dec 2025

Emergent Software | Minneapolis, MN

- Architected cloud-native microservices-based platform using **ASP.NET Core (.NET 6)**, **C#**, **React**, **TypeScript**, **Umbraco CMS**, and **Vendr Commerce**, delivering a unified multi-resort ecommerce experience across PA, OH, WI, and TX.
- Designed high-availability **Azure** infrastructure with Azure Front Door geo-routing, autoscaling App Services, **Redis** distributed cache, **CDN**, **Azure SQL**, **Microsoft SQL Server**, and multi-region failover, scaling traffic from 3,580 to 15,000+ RPM (50x) with zero downtime during national campaigns.
- Built modular React frontend with **Next.js**, **TailwindCSS**, and custom Umbraco Block Editor components, enabling **SSR**, mobile-first **UI/UX**, **SEO** optimization, and elimination of third-party booking redirects.
- Implemented event-driven multilingual automation using **n8n** + **Azure Cognitive Translator** + **Webhooks** + background workers (**Hangfire**), reducing manual localization effort while supporting full-site translation.
- Developed **RESTful microservices** and **async APIs** for cart, content, localization, and personalization, integrating **Redis caching**, message queues, and **EF Core pipelines**; migrated 600+ pages (thousands of nodes) via automated Umbraco import tools.
- Established production-grade **DevOps** and observability with **Azure DevOps CI/CD**, **Docker**, **Terraform/ARM**, Application Insights, **Azure Monitor**, **Serilog**, health checks, load testing (**JMeter/Azure Load Testing**), and automated alerting.
- Delivered enterprise security and compliance using **Azure WAF**, **GDPR tooling (Cookiebot)**, **reCAPTCHA**, **Azure AD B2C**, achieving **WCAG** accessibility standards and earning an Umbraco Award for Best Editing Experience.

ASP.NET Core (.NET 6), C#, React, TypeScript, Next.js, Umbraco CMS (v9+), Vendr Commerce, Microservices Architecture, REST APIs, Entity Framework Core, Azure App Services, Azure Front Door, Azure Load Balancer, Azure Traffic Manager, Azure SQL Database, Azure Blob Storage, Azure CDN, Azure Cache for Redis, Docker, Terraform, ARM Templates, Azure DevOps Pipelines, CI/CD, n8n, Azure Cognitive Translator, Hangfire, Webhooks, Message Queues, Application Insights, Azure Monitor, Log Analytics, Serilog, Health Checks Middleware, JMeter, Azure Load Testing, TailwindCSS, WCAG 2.1 Accessibility, React Block Editor Extensions, Cookiebot, Azure WAF, reCAPTCHA, Azure AD B2C, GDPR Compliance, SSR, Lighthouse Optimization, Redis Caching, CDN Edge Caching, Async APIs, Background Workers, Multilingual Automation, Content Migration Pipelines

Senior Full Stack Software Engineer (.NET/Angular)

Feb 2019 - Apr 2022

BairesDev | San Francisco, CA

- Architected and delivered a dynamic form configuration system using **ASP.NET MVC(.NET 6)**, **Angular**, and **JSON** Schema-driven APIs, enabling administrators to create, modify, and version custom form fields and layouts, which reduced client onboarding time and eliminated hard-coded form dependencies.
- Implemented high-throughput event-driven data processing with **Apache Kafka** integrated into **.NET Core** background services, enabling real-time ingestion of call events, lead submissions, and form updates, resulting in near-real-time analytics delivery and improved system scalability under peak traffic.
- Optimized large-scale data storage and retrieval using **Apache Cassandra** for write-heavy workloads (call logs, form submissions, event streams), significantly reducing query latency and removing relational database bottlenecks under millions of daily events.
- Built a modular Angular-based admin interface leveraging **Angular Reactive Forms**, **Angular CDK**, and dynamic component rendering, allowing real-time schema-driven UI updates synchronized with **backend APIs**, which improved admin productivity and reduced form-related support tickets.
- Integrated real-time communication and system synchronization using **SignalR** with ASP.NET Core, enabling instant propagation of form changes and live lead updates across services, delivering sub-second UI refresh times and improved user experience.
- Improved overall system performance and reliability by containerizing services with **Docker**, deploying scalable workloads via **Kubernetes**, and tuning Kafka consumers and Cassandra partition strategies, achieving higher throughput, lower processing latency, and stable performance during traffic spikes.

ASP.NET Core (.NET 6), C#, Angular, Angular Reactive Forms, Angular CDK, JSON Schema, SignalR, Apache Kafka, Kafka Consumers, Event-Driven Architecture, Apache Cassandra, Distributed NoSQL Databases, High-Throughput Data Pipelines, Real-Time Analytics Systems, RESTful APIs, Microservices Architecture, Docker, Kubernetes, Containerized Deployments, CI/CD Pipelines, Performance Optimization, Low-Latency Systems, Database Partitioning, Query Optimization, Scalability Engineering, CRM Integrations, Telephony Integrations, Dynamic Number Insertion (DNI), Real-Time Data Synchronization, Cloud-Native Application Design

Full Stack Software Engineer (.NET/React)

Jun 2015 - Jan 2019

Svitla Systems | Corte Madera, CA

- Architected and delivered a cross-platform social networking platform using **Flutter (Dart)** for mobile and **React.js + Next.js + TypeScript + Redux Toolkit** for web, backed by **ASP.NET Core Web API**, achieving production launch within 6 months and increasing early user engagement through a stable multi-platform release.
- Built high-performance backend services with **.NET Core**, **Entity Framework Core**, **Dapper**, **MediatR (CQRS)** and optimized **MySQL queries** using index tuning, query refactoring, **Redis caching**, and **Amazon RDS**, reducing **API response** latency and eliminating database bottlenecks under concurrent user load.
- Implemented scalable cloud infrastructure on **AWS (EC2, ALB, S3, CloudFront, Lambda, SNS, SQS, RDS, CloudWatch)** using **Docker**, **GitHub Actions CI/CD**, and **Terraform**, improving deployment reliability and enabling horizontal scaling that supported growing traffic without downtime.
- Developed secure authentication and authorization flows using **JWT**, **OAuth2**, **ASP.NET Identity**, and **OTP verification** via **AWS SNS**, while enforcing role-based access control, resulting in hardened platform security and zero critical auth vulnerabilities in production.
- Integrated real-time and asynchronous processing using **SignalR**, **Redis Pub/Sub**, **AWS SQS/SNS**, and background jobs with **Hangfire**, improving feed delivery and notification throughput while maintaining sub-second response times during peak activity.

- Implemented subscription and in-app purchase pipelines using **Apple StoreKit**, **Google Billing Library**, **Stripe webhooks**, and backend entitlement services in .NET, increasing monetization readiness and enabling premium feature rollout with measurable conversion growth.

Flutter (Dart), React.js, Next.js, TypeScript, Redux Toolkit, ASP.NET Core, .NET Web API, RESTful APIs, gRPC, Entity Framework Core, Dapper, MediatR (CQRS), SignalR, MySQL, Amazon RDS, Redis, Elasticsearch/OpenSearch, AWS EC2, ALB, S3, CloudFront, Lambda, SNS, SQS, CloudWatch, Docker, Kubernetes (EKS), Terraform, GitHub Actions CI/CD, JWT, OAuth2, ASP.NET Identity, Stripe, Apple StoreKit, Google Billing, Firebase Analytics, Crashlytics, WebSockets, Hangfire, FFmpeg, CDN optimization, database indexing, query optimization, caching strategies, connection pooling, horizontal scaling, load balancing, high-performance APIs, latency reduction, bottleneck analysis, microservices architecture, distributed systems, event-driven architecture, secure authentication, role-based access control (RBAC), Agile/Scrum.

EDUCATION

Bachelor's Degree in Computer Science
University of Texas at Austin

Aug 2011 - May 2015

PROJECT

Private Family Social Networking Platform

<https://app.yourstotell.com>

Developed an ads-free, cross-platform social platform enabling families to securely connect, share multimedia stories, manage relationships via Family Tree graphs, and control content access through role-based privacy settings. Delivered real-time feeds, tiered album management, in-app subscriptions, and admin moderation tools on scalable cloud infrastructure with low-latency APIs and optimized databases to support growing user communities.

Call Tracking & Lead Management Platform

<http://matelso.com>

Developed a cloud-based call tracking and lead management platform that captures, processes, and analyzes customer interactions across multiple channels. The system enables businesses to dynamically create and customize contact forms, track inbound calls with real-time attribution, manage leads from multiple entry points, and synchronize interaction data across analytics and CRM systems. It provides real-time visibility into call activity, form submissions, and customer journeys, supports multichannel attribution and conversation tracking, and allows administrators to configure communication workflows and form layouts without code changes, improving marketing performance and customer engagement at scale.